on organisms and human health. Along with atmospheric factors (humidity, precipitation, wind, air currents, altitude) prevailing at a place and time, its effects can be far reaching and catastrophic.

Air pollutants can be

- discharge of dusts or particulate matter (PM : 2.5 μm)
- discharge of gases (SO_2, NO_2, CO, CO_2)

Carbon monoxide (CO) is produced mainly due to incomplete combustion of fossil fuels. Automobiles are major causes of CO pollution in large cities and towns Automobile exhausts, fumes from factories, emission from power plants, forest fires and burning of fire-wood contribute to CO pollution.

With rapid urbanization, major amount of carbon dioxide and sulphur dioxide (SO_2) is released in the atmosphere. From automobiles, aeroplanes, power plants and other human activities that involving the burning of fossil fuels (coal, oil etc.,) CO_2 is the main pollutant that is leading to **global warming**.

Nitrogen oxides are also major air pollutants. Fossil fuel combustion and automobiles exhausts are the source of nitrogen oxides. Sulphur dioxide and nitrogen oxides are the major causes of acid rain.

Particulate matters are tiny particles of solid matter suspended in a gas or liquid. Combustion of fossil fuels, fly ash produced in thermal power plants, forest fires, asbestos mining units, cement factories are the main sources of particulate matter pollution.

13.2.1 Sources

The main sources of air pollution **(Fig 13.1)** are:

- **Transport sources** – cars, buses, airplanes, trucks, trains
- **Stationary sources** – power plants, incinerators, oil refineries, industrial facilities, and factories
- **Area sources** – agricultural - wood / stubble burning, fireplaces
- **Natural sources** – wind-blown dust, wildfires, volcanoes.

13.2.2 Effects of Air Pollution

- Affects all organisms as they depend on the atmosphere for respiration.
- Causes irritation in the throat, nose, lungs and eyes. It causes breathing problems and aggravates existing health conditions such as emphysema and asthma.
- Contaminated air reduces the body's defense mechanism and decreases the body's capacity to fight other infections in the respiratory system.
- Frequent exposure to polluted air increases the risk of cardiovascular diseases. Breathing air that is filled with fine particulate matter can induce hardening of the arteries, triggering cardiac arrhythmia or even a heart attack.
- People who exercise outdoors can sometimes be susceptible to adverse effects of air pollution because it involves deeper and faster breathing. Hence it is advisable to walk or jog in the mornings in places with ample tree cover.
- Gas leaks can be lethal or affect the quality of air in the affected area.
- CO in the atmosphere interferes with O_2 transport since haemoglobin has greater affinity for carbon monoxide. At low concentration it causes headache and blurred vision. In higher concentration, it can lead to coma and death.



Sameer, an App provides hourly updates on the National Air Quality Index (AQI) published by CPCB.

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